

Nathaniel Wihardjo

SENIOR SOFTWARE ENGINEER

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Experience

Spotify

London, United Kingdom

SENIOR SOFTWARE ENGINEER

Oct. 2022 - Present

- Drove a project across 3 product areas and implemented the C++ algorithm that detects ads inserted into podcast episodes, unlocking chapters, ad skipping, and transcript features for 49% of podcast streams and raising chapter usage by 58%
- Conceived and shipped the first LLM-powered on-call assistant that automatically diagnoses backend incidents, cutting root-cause analysis time from 45 to 4 minutes. Since evolved and adopted as company-wide standard, handling 600+ incidents weekly
- Led Offline Backup feature from hackathon prototype to a full Premium tier launch, letting listeners play cached music without a connection. It reached 13.7M users to date and serves 2M hours of offline listening weekly
- Redesigned the C++ offline system serving 20% of listening time across 310M devices to be content-agnostic, then partnered across teams to build Spotify's first music video and partial download features, growing music video consumption by 9%
- Developed automatic video-autoplay logic into the app's playback engine, boosting autoplay consumption by 12% across 55.7M exposed users
- Designed and shipped a cross-device state-synchronization system that reconciles play history for 600M+ listeners in near real time, cutting repeated tracks during shuffle playback by 3.6%

Macquarie Group

London, United Kingdom

MANAGER, SOFTWARE ENGINEER

Aug. 2019 - Oct. 2022

- Led a 3-engineer team to re-architect in-house application as Python/Flask microservices, improved speed of delivery by 50%, and reduced labour costs by 80%
- Developed netback pricing model in C++ for LNG trade valuation, exposed via GUI and API, generating 10% of European energy business profit
- Built a parallel news-driven oil commodity trading strategy in Python, generating £200K+ in annualised P&L
- Replaced overnight batch and ad-hoc risk processing with real-time risk reports across 3 trading desks, partnering with traders on the firm's global C++ trading system
- Mentored 4 junior developers on career growth, coding standard and best practices, and financial products understanding

TECHNOLOGY INTERN

Jun. 2018 - Aug. 2018

- Built a big-data platform with Cloudera cluster management on AWS EC2 instances for robust and scalable system
- Resolved incompatibility between Apache Impala and Kerberos protocol with bash scripting and gRPC protocol
- Delivered data analysis to the Energy Market Access trading desk

Education

Hong Kong University of Science and Technology

Hong Kong S.A.R

B.ENG IN CHEMICAL ENGINEERING, MINOR IN INFORMATION TECHNOLOGY - FIRST CLASS HONOURS

2015 - 2019

- Peer-reviewed publication: A.K. Oktavius *et al.*, "Fully-Conformable Porous Polyethylene Nanofilm Sweat Sensor for Sports Fatigue" IEEE Sensors Journal 21.7 (2021): 8861-8867

University College London

London, United Kingdom

AFFILIATE STUDENT, CHEMICAL ENGINEERING - FIRST CLASS HONOURS

2017 - 2018

Highlight Projects

Spaghetti Search

Go, BadgerDB

CONCURRENT AND SCALABLE SEARCH ENGINE BUILT ON GOLANG

[Code here](#)

- Utilised Topic-Sensitive PageRank and Vector-Space Model algorithm for ranking on a NoSQL database

Trading Bot

pytorch, numpy

A REINFORCEMENT LEARNING MINUTE-BASED FINANCIAL TRADING AGENT ON HKEX

[Code here](#)

- Developed base code for trading bot using Deep Direct Reinforcement Learning model

Skills

Language C++20, Python, Java, Bash

Technologies Bazel, gRPC/Protobuf, Boost, LevelDB, GCP (Pub/Sub, GCS, BigQuery), Grafana

Domains Playback orchestration, offline/caching, A/B experimentation